

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1949

Luray Hydro Station 138 kV
December 26, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1949. The Interconnection Customer proposes a 92.9 MW solar photovoltaic facility interconnecting at the Luray Hydro Station 138 kV substation in the Virginia area. The request was received on December 01, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 456 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,541,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Templeton Substation to Buckhorn Substation 138 KV	Springfield Power Llc to Edinburg 138 KV	115.9	172	12.0
Roslyn Substation to Buckhorn 138 KV L	Springfield Power Llc to Edinburg 138 KV Line	113.9	177	12.0
Red Tail to Templeton 138 KV	Edinburg to Springfield Power Llc 138 KV L	106.3	186	12.0
Lewisurg to S Jordan 138 KV Line	Detour to Fort Benton 138 KV	105.7	343	5.3
Siloam Springs to Lewisburg 138 KV	Fort Benton to Detour 138 KV Line	100.7	437	7.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Buckhorn to Templeton Substation 138 KV	Rebuild line with higher-capacity conductor	\$5,829,000	6% (\$370,000)
Roslyn Substation to Buckhorn 138 KV L	Rebuild line with higher-capacity conductor	\$8,528,000	6% (\$542,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Red Tail to Templeton 138 KV	Reconductor line	\$3,302,000	6% (\$210,000)
Lewisburg to S Jordan 138 KV	Reconductor line	\$2,598,000	43% (\$1,125,000)
Siloam Springs to Lewisburg 138 KV	Reconductor line	\$2,219,000	13% (\$294,000)
	Total allocated to project		\$2,541,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.